

and Li (2012) replicated smart beta strategies as closely as possible using conventionally available products. Here are their annual portfolio turnover figures: S&P 500 Indexing 6.7%, Fundamental Indexing 14%, Equal Weight Indexing 22.9%, and Minimum Variance Indexing 49.2%. Low volatility/minimum variance can also have high tracking error, which measures how much a strategy return differs from its benchmark return. While low volatility and low beta strategies show some promise in academic tests, the problems of implementation with respect to portfolio turnover and tracking error may eliminate much of their advantage.

Sector concentration can be another problem for low volatility strategies. The S&P 500 Low Volatility Index invests in the 100 stocks of the S&P 500 Index having the lowest volatility over the preceding 12 months. It does not constrain sector weights, which can result in huge sector concentrations that cause large tracking errors. Right now, for example, 62% of the S&P Low Volatility Index is in three sectors, and 76% is in four sectors. There have been times in which over two-thirds of the index has been in only two sectors. [Figure 6.4](#) shows how well PowerShares S&P 500 Low Volatility (SPLV), based on the S&P Low Volatility Index, matches up with just the defensive, low volatility SPDR Consumer Staples (XLP) sector.